

EMBEDDED DEVICE FOR ALZHEIMER'S DISEASE PATIENTS

A PROJECT REPORT

submitted in partial fulfillment of the requirements

for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ELECTRICAL AND ELECTRONICS ENGINEERING

by

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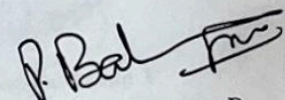
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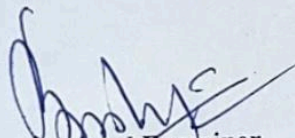
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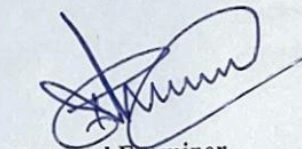
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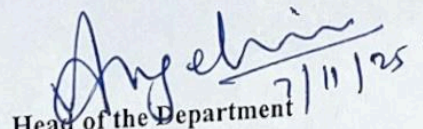

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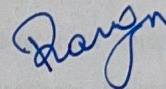

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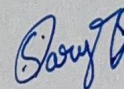
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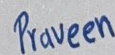
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ABSTRACT

Alzheimer's disease is a progressive neurologic disorder that causes memory loss, disorientation, and diminished capacity to carry out activities of daily living, usually resulting in safety hazards for the patients. This project considers the task of designing and implementing an embedded assistive device for monitoring and assisting Alzheimer's patients in their everyday life. The system to be designed incorporates various sensors: the NEO-6M GPS module for tracking location in real time, a Pulse Sensor for monitoring heart rate throughout the day, the MPU-6050 sensor for detecting falls, and a GSR sensor to analyze the stress level of the patient. The ESP32 microcontroller serves as the brain unit, handling sensor data and communication. A speaker module is utilized for playing pre-recorded voice reminders for medication or daily activities. The product increases patient independence and support from caregivers, along with enhancing patient safety, through real-time monitoring and alerts, presenting a low-cost, easy-to-use, and reliable solution for Alzheimer's care.

TABLE OF CONTENTS

LIST OF TABLES.....	vii
LIST OF FIGURES.....	vii
ABBREVIATIONS AND.....	ix
CHAPTER I.....	1
1. INTRODUCTION.....	1
1.1 INTRODUCTION.....	1
1.1.1 Motivation.....	1
1.1.2 Objectives.....	1
1.1.3 Scope of the Work.....	2
1.2 ORGANIZATION OF THESIS.....	2
CHAPTER II.....	4
2. PROJECT DESCRIPTION.....	4
2.1 OVERVIEW OF PROJECT.....	4
2.2. MODULES OF THE PROJECT.....	4
2.1.1 Module 1.....	4
2.1.2 Module 2.....	5
2.1.3 Module 3.....	5
2.1.4 Module 4.....	5
2.1.5 Module 5.....	5
2.1.6 Module 6.....	5
2.2 TASKS AND MILESTONES.....	6
CHAPTER III.....	7
3. DESIGN OF EMBEDDED DEVICE FOR ALZHEIMER’S DISEASE PATIENTS.....	7
3.1 DESIGN APPROACH.....	7
3.1.1 Codes and Standards.....	7
3.1.2 Realistic Constraints.....	7
3.1.3 Alternatives and Tradeoffs.....	8
3.2 DESIGN SPECIFICATIONS.....	9
CHAPTER IV.....	11
4. PROJECT DEMONSTRATION.....	11
4.1 INTRODUCTION.....	11
4.2 ANALYTICAL RESULTS.....	11
4.3 HARDWARE RESULTS.....	12

CHAPTER V.....	15
5. CONCLUSION.....	15
5.1 COST ANALYSIS.....	15
5.2 SCOPE OF WORK.....	16
5.3 SUMMARY.....	16
REFERENCES.....	18
PUBLICATIONS BASED ON THESIS.....	19
CURRICULUM VITAE.....	20

LIST OF TABLES

Table 1: Task Progress and Milestones	6
Table 2: Alternative Considerations	8
Table 3: Design Specifications	9
Table 4: Cost analysis and approximation	15

LIST OF FIGURES

Figure 1: Hardware connection Diagram	10
Figure 2: Hardware result readings of real time data	13
Figure 3: Reading of real time data Blynk Website	14
Figure 4: Real time data alert Blynk website for mail	14

ABBREVIATIONS AND NOMENCLATURE

BPM	Beats Per Minute
GPS	Global Positioning Service
GSR	Galvanic Skin Resistance
UART	Universal Asynchronous Receiver/Transmitter
I2C	Inter-Integrated Circuit

CHAPTER I

1. INTRODUCTION

1.1 INTRODUCTION

Alzheimer's disease is a progressive neurological disorder that affects memory, reasoning, and daily functioning, making patients increasingly dependent on caregivers. As the disease progresses, patients face challenges such as disorientation, forgetfulness, and health complications that can lead to safety risks. To address these issues, this project focuses on developing an embedded monitoring and assistance device for Alzheimer's patients. The system provides real-time health monitoring, location tracking, and audio reminders to help patients maintain their routines while ensuring caregiver awareness and support.

1.1.1 Motivation

The motivation for developing this project arises from the need to improve patient safety, independence, and caregiver support for individuals suffering from Alzheimer's. Patients often forget daily tasks, medications, or wander from home, creating distress for families. This device uses embedded technology and sensors to continuously monitor vital parameters such as heart rate, stress level, body movement, and location. By integrating these with real-time alerts and task reminders, the project aims to create an affordable, reliable solution that enhances quality of life for patients and reduces caregiver burden.

1.1.2 Objectives

The main objectives of this project are:

- To design an embedded system that continuously monitors vital signs and movement of Alzheimer's patients.
- To use GPS (NEO-6M) for real-time location tracking and safety monitoring.

- To measure heart rate using a Pulse Sensor and stress level using a GSR sensor.
- To detect falls using the MPU-6050 motion sensor.
- To remind patients of daily tasks and medications using pre-recorded voice prompts through a DFPlayer speaker.
- To integrate all sensors using an ESP32 microcontroller for data processing and communication with caregivers.

1.1.3 Scope of the Work

The scope of this project includes the design, development, and testing of an embedded device capable of real-time monitoring and alert generation. The system covers:

- Health monitoring through multiple sensors (Pulse, GSR, MPU-6050).
- Location tracking via GPS.
- Task and medication reminders through audio alerts.
- Data processing and control via ESP32.
- Communication with caregiver devices for alerts.

The project is limited to non-clinical health assistance and does not replace professional medical care. Future improvements may include cloud data storage, mobile app integration, and AI-based health prediction for enhanced functionality as well using of ML modelling can be integrated in the system for storing and as well predicting of stress levels of the user with which the improvements in the patients disease and as well proper monitoring of the patient along his life for medications to be followed.

1.2 ORGANIZATION OF THESIS

This thesis is organized into five chapters, each describing a specific phase of the project.

- **Chapter 1: Introduction:**
Provides an overview of the project, including motivation, objectives, scope, and problem definition related to developing an embedded assistive device for Alzheimer's patients.

- **Chapter 2 :Project Description:**
Describes the overall concept and working of the system, detailing various modules such as GPS tracking, heart rate, fall detection, stress monitoring, and audio reminder features.
- **Chapter 3: Project Design:**
Explains the design methodology, hardware and software specifications, and the integration of different components used in the embedded system.
- **Chapter 4: Project Demonstration:**
Presents the implementation, analytical results, hardware testing, and simulation outcomes of the developed prototype.
- **Chapter 5 :Conclusion and Future Scope:**
Summarizes the project outcomes, cost analysis, scope of work, and suggests future enhancements like mobile app connectivity and AI-based monitoring.

CHAPTER II

2. PROJECT DESCRIPTION

2.1 OVERVIEW OF PROJECT

Alzheimer's disease causes memory loss, confusion, and difficulty in performing daily activities, which makes constant monitoring essential for patient safety. The proposed system is an embedded device for Alzheimer's patients which is designed to provide continuous health and safety monitoring using various sensors integrated with an ESP32 microcontroller. The device can track the patient's location, detect falls, monitor heart rate and stress levels, and provide audio reminders for daily tasks or medicine intake.

This system assists caregivers by sending real-time alerts and updates, ensuring better supervision even from a distance. The device aims to improve the independence, safety, and well-being of Alzheimer's patients while reducing the emotional and physical stress on caregivers.

2.2 MODULES OF THE PROJECT

The proposed system consists of several functional modules, each responsible for monitoring or assisting a specific aspect of the patient's daily life. These modules are integrated and controlled using the ESP32 microcontroller, which processes sensor data and manages output functions such as voice alerts.

2.2.1 Module 1 – GPS Tracking Module (NEO-6M)

This module uses the NEO-6M GPS sensor to provide real-time location tracking of the patient. It continuously updates the coordinates and sends alerts to caregivers if the patient moves beyond a defined area. This ensures patient safety and helps in quick recovery in case of wandering or disorientation — a common symptom in Alzheimer's patients.

2.2.2 Module 2 – Heart Rate Monitoring Module (Pulse Sensor)

The Pulse Sensor measures the patient's heart rate in beats per minute (BPM). Abnormal readings can indicate stress, anxiety, or a possible health emergency. This data can be logged or transmitted to caregivers to help them monitor the patient's well-being.

2.2.3 Module 3 – Fall Detection Module (MPU-6050)

The MPU-6050 sensor detects motion, acceleration, and orientation. Sudden abnormal movement or impact is identified as a fall, triggering an alert through the microcontroller. This allows caregivers to respond immediately to prevent injury or provide assistance.

2.2.4 Module 4 – Stress Monitoring Module (GSR Sensor)

The GSR (Galvanic Skin Response) sensor measures skin conductivity, which varies with stress or anxiety levels. This helps in assessing the patient's emotional state and detecting possible discomfort or agitation.

2.2.5 Module 5 – Voice Reminder Module (Speaker + DFPlayer Mini)

This module uses a speaker and DFPlayer Mini to play pre-recorded audio messages that remind the patient to take medicines, drink water, or perform daily tasks. These reminders help maintain the patient's routine and reduce dependency on caregivers.

2.2.6 Module 6 – Central Processing Unit (ESP32 Microcontroller)

The ESP32 acts as the core controller of the system. It collects data from all sensors, processes it, and manages outputs such as alerts and reminders. The ESP32 is chosen for its low power consumption, Wi-Fi/Bluetooth connectivity, and ability to handle multiple sensors efficiently.

2.2 TASKS AND MILESTONES

Task / Milestone	Description	Status / Expected Output
Requirement Analysis	Study of patient needs, caregiver requirements, and system objectives along with research papers for identifying research gaps and scope for development.	Completed – Defined project goals.
Hardware Selection	Selection of ESP32, GPS, Pulse, MPU-6050, and GSR sensors, audio module-DF mini player	Completed – Components finalized.
Circuit Design & Integration	Connecting all sensors with ESP32 and ensuring compatibility.	Completed – Working prototype built.
Programming & Testing	Writing code for data collection, fall detection, and voice alerts.	completed – System testing and results
Data Monitoring & Alerts	Integration of location tracking and health monitoring features.	Completed – Real-time alerts upon trigger and time.
Final Implementation	Deploying the system for real-time patient monitoring.	To Be Completed – Ready for demonstration.

Table 1 : Task Progress and Milestones

CHAPTER III

3. DESIGN OF EMBEDDED DEVICE FOR ALZHEIMER'S DISEASE PATIENTS

3.1 DESIGN APPROACH

The proposed system is an embedded device designed to assist Alzheimer's patients by monitoring their vital parameters, detecting abnormal events, and providing timely task reminders. The design integrates multiple sensors GPS NEO-6M, Pulse Sensor, MPU-6050, and GSR Sensor with an ESP32 microcontroller as the central control unit. A speaker module (DFPlayer Mini with speaker) is used to play pre-recorded voice messages for reminders and alerts.

The system continuously collects data from sensors, processes it through ESP32, and provides audio or digital alerts to caregivers when abnormal readings or movements are detected. The design aims to achieve low cost, compact size, real-time monitoring, and user-friendly operation, suitable for home-based patient care.

3.1.1 Codes and Standards

The design and development of this project adhere to standard embedded system and biomedical electronics practices:

- **Embedded Hardware Standards:** Use of regulated 3.3V and 5V logic levels for microcontrollers and sensors.
- **Communication Standards:** UART protocol for GPS, serial communication for DFPlayer Mini, and I2C for MPU-6050.
- **Safety Standards:** Low-voltage DC power supply ensures user safety.
- **Coding Standards:** Follows modular programming using C (Arduino IDE), ensuring readability and easy debugging.
- **Data Accuracy:** Sensors calibrated as per manufacturer datasheets for reliable physiological readings.
- **Ethical Standards:** Designed for assistance and not for clinical diagnosis, maintaining patient data privacy.

3.1.2 Realistic Constraints

The project design is subject to several real-world constraints:

- **Power Constraint:** Operates on limited battery capacity, so energy-efficient components (ESP32) are used.
- **Cost Constraint:** Designed to be low-cost using readily available sensors and modules.
- **Size Constraint:** Compact design to ensure comfort when used as a wearable or portable device.
- **Environmental Constraint:** Sensors may be affected by ambient temperature, humidity, or motion noise.
- **User Constraint:** Must be easy to use for elderly patients and caregivers with minimal technical knowledge.
- **Connectivity Constraint:** GPS and Wi-Fi performance may vary with signal strength or indoor usage.

3.1.3 Alternatives and Trade-offs

During development, several alternatives were considered for sensors and components:

Component	Alternative Considered	Trade-off Decision
ESP32	Arduino Uno / NodeMCU	ESP32 chosen for built-in Wi-Fi, Bluetooth, and dual-core performance.
GPS NEO-6M	SIM808 (GSM + GPS)	NEO-6M selected for lower cost and easy integration.
Pulse Sensor	MAX30100 / MAX30102	Pulse Sensor chosen for simplicity and low power consumption.
MPU-6050	ADXL345	MPU-6050 chosen for combined accelerometer, gyroscope, and magnetometer.
GSR Sensor	ECG Sensor	GSR used for stress detection — cost-effective and non-invasive.
DFPlayer Mini + Speaker	Buzzer / OLED Display	DF Player is chosen for voice reminders, providing better user communication.

Table 2: Alternative Considerations

These trade-offs ensured that the final design remained cost-effective, power-efficient, and user-friendly, without compromising functionality.

3.2 DESIGN SPECIFICATIONS

3.2.1 Hardware Design Table

Component	Function / Role	Interface Used	Power Supply	Remarks
ESP32 Microcontroller	Central control unit; processes sensor data and manages communication.	GPIO, UART, I ² C	3.3V	Core processing unit.
NEO-6M GPS Module	Tracks real-time patient location.	UART (TX/RX)	3.3V–5V	Sends latitude & longitude data.
Pulse Sensor	Measures heart rate (BPM).	Analog Input	3.3V	Detects pulse using photoplethysmography.
MPU-6050 Sensor	Detects falls using acceleration and orientation data.	I ² C	3.3V	Combines accelerometer, gyroscope, and magnetometer.
GSR Sensor	Measures skin resistance to determine stress level.	Analog Input	3.3V–5V	Indicates emotional state.
DFPlayer Mini + Speaker	Plays voice reminders for daily tasks and medicine alerts.	Serial	5V	Improves patient guidance.
Power Supply (Battery)	Provides required operating voltage for all components.	-	5V	Portable and rechargeable power source.

Table 3: Design Specifications

3.2.2 System Explanation

- **Input Stage:** Sensors (Pulse, GSR, MPU-6050, GPS) collect data from the patient's body and environment.
- **Processing Stage:** ESP32 receives data, filters noise, detects abnormal events (like fall or stress), and triggers alerts.
- **Output Stage:** DFPlayer Mini plays voice reminders or alert messages through the speaker.
- **Communication Stage:** Data can be sent wirelessly (if enabled) to caregivers' devices for monitoring.

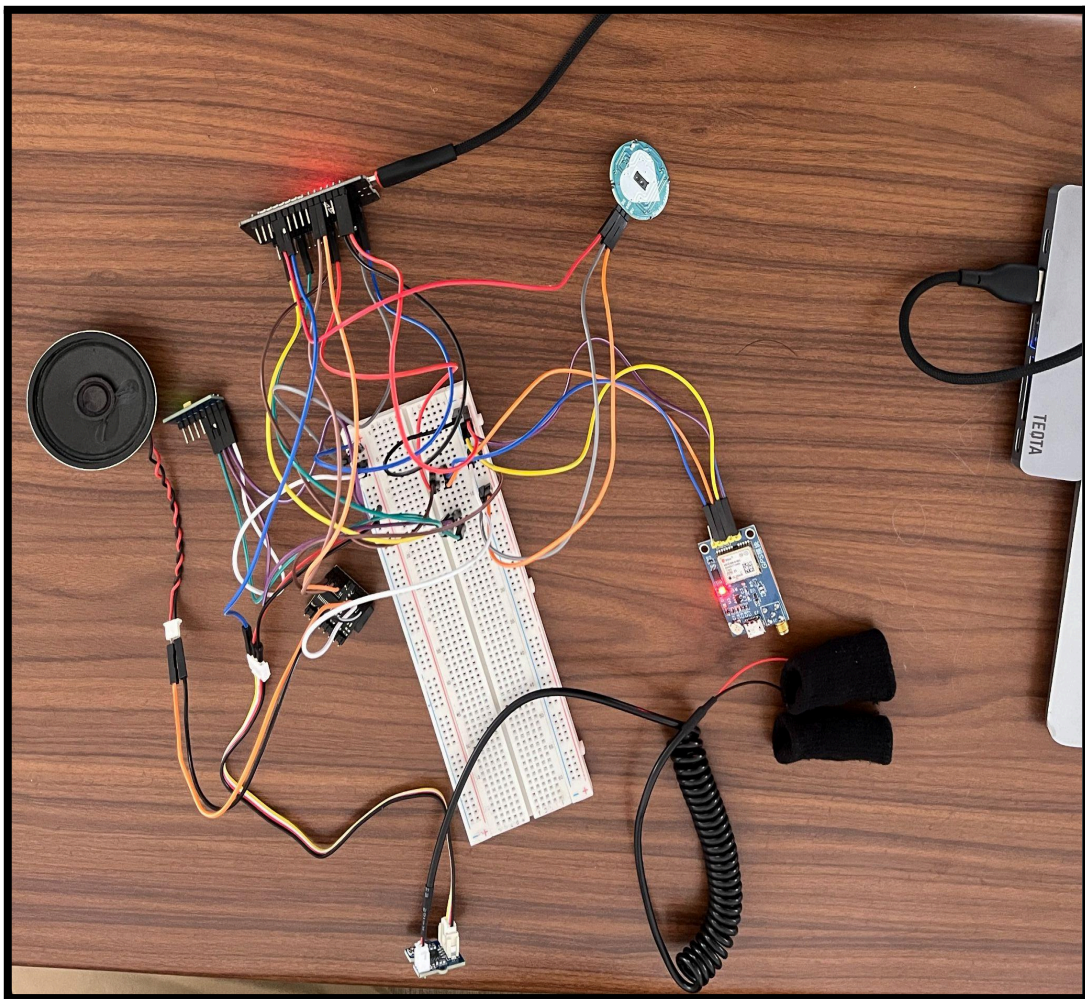


Figure 1: Hardware connection Diagram

CHAPTER IV

4. PROJECT DEMONSTRATION

4.1 INTRODUCTION

The proposed project, Embedded Device for Alzheimer's Patients, aims to provide an assistive monitoring system that ensures safety, health tracking, and daily support for individuals suffering from Alzheimer's disease. The device integrates multiple sensors: GPS NEO-6M for location tracking, Pulse Sensor for heart rate measurement, MPU-6050 for fall detection, and GSR Sensor for stress monitoring — all controlled by an ESP32 microcontroller.

Additionally, the device includes a DFPlayer Mini and speaker to deliver pre-recorded voice reminders for medication, hydration, and daily tasks. The system functions in real-time, continuously gathering physiological and movement data, analyzing it, and providing alerts to caregivers when abnormalities occur.

This demonstration focuses on how each module operates, how the integrated system behaves in real-time, and the results obtained from both hardware implementation and simulation testing.

4.2 ANALYTICAL RESULTS

The analytical results demonstrate the effective integration and functionality of each module within the embedded system.

- The GPS module successfully provides accurate latitude and longitude coordinates, enabling caregivers to locate the patient in real-time.
- The Pulse Sensor records heart rate in beats per minute (BPM). During testing, normal readings between 60-180 BPM, showing stable operation.

- The MPU-6050 sensor accurately detects falls by sensing sudden acceleration and orientation changes, immediately triggering an alert.
- The GSR sensor measures variations in skin resistance to indicate stress levels; high stress corresponds to higher conductivity.
- The ESP32 processes all data efficiently and triggers appropriate actions such as playing voice messages or sending alerts.
- The DFPlayer Mini and speaker output clear voice reminders like *“It’s time to take your medicine”* or *“Please drink water.”*

These results confirm that the device operates reliably in monitoring physiological parameters, detecting critical events, and assisting patients through audio communication.

4.3 HARDWARE RESULTS

The hardware prototype was built and tested using all modules connected to the ESP32 microcontroller.

Observations:

- **GPS Module (NEO-6M):**
 - Provided real-time tracking data via serial monitor.
 - Location updates were accurate within a 5–10 meter range.
- **Pulse Sensor:**
 - Output displayed on serial monitor as analog values converted to BPM.
 - Detected live pulse changes when the user's finger was placed on the sensor.
- **MPU-6050 (Fall Detection):**
 - Detected abrupt accelerations and orientation changes.
 - Triggered alert condition when a fall-like movement was simulated.
- **GSR Sensor:**
 - Measured skin conductivity variations indicating stress or calm state.
 - Higher values corresponded to increased stress levels.

- **DFPlayer Mini + Speaker:**
 - Played pre-recorded reminders clearly and instantly upon trigger.
- **ESP32 Controller:**
 - Processed all sensor inputs efficiently.
 - Used serial monitor and LED indicators for debugging and verification.

Conclusion:

All modules worked in synchronization, and the embedded system provided a smooth, stable, and real-time response. The hardware demonstration verified the successful integration and accuracy of the proposed design

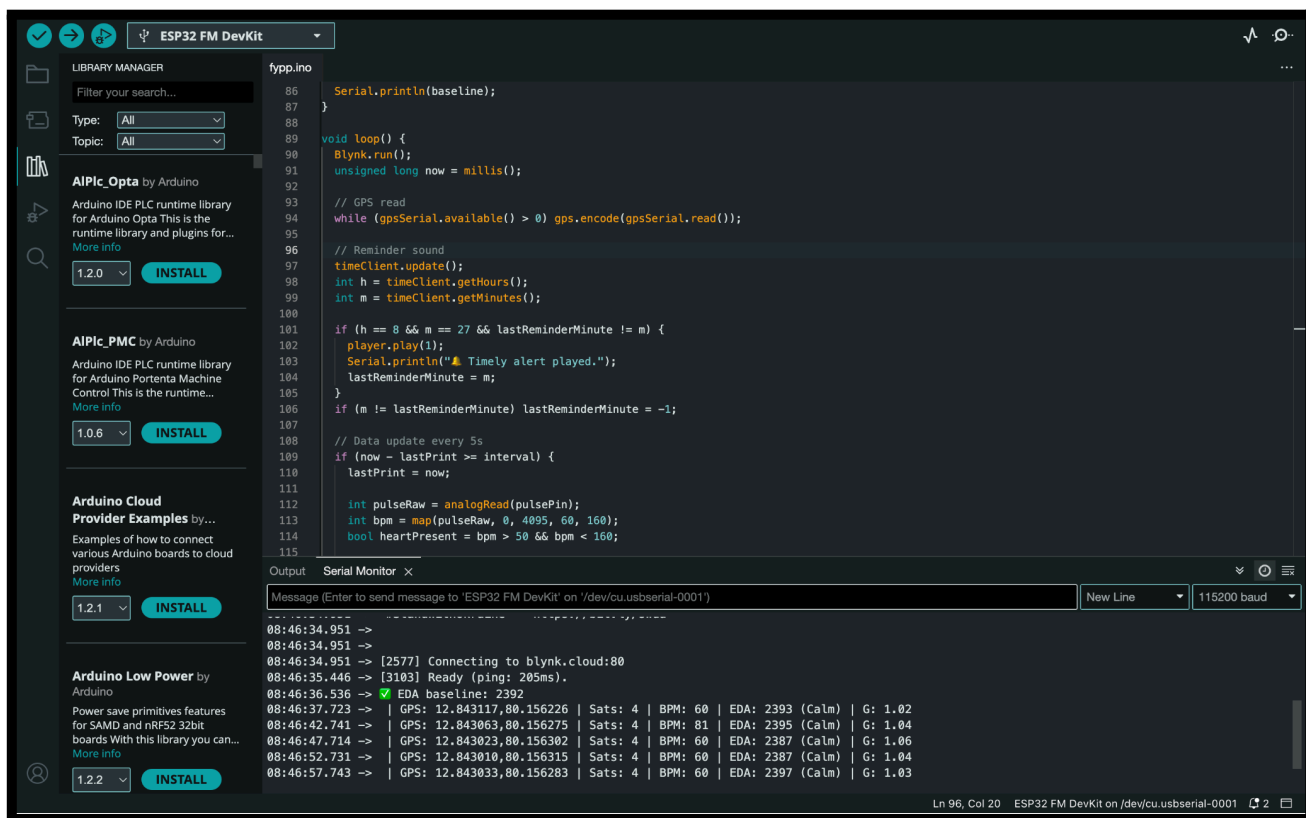


Figure 2:Hardware result readings of real time data

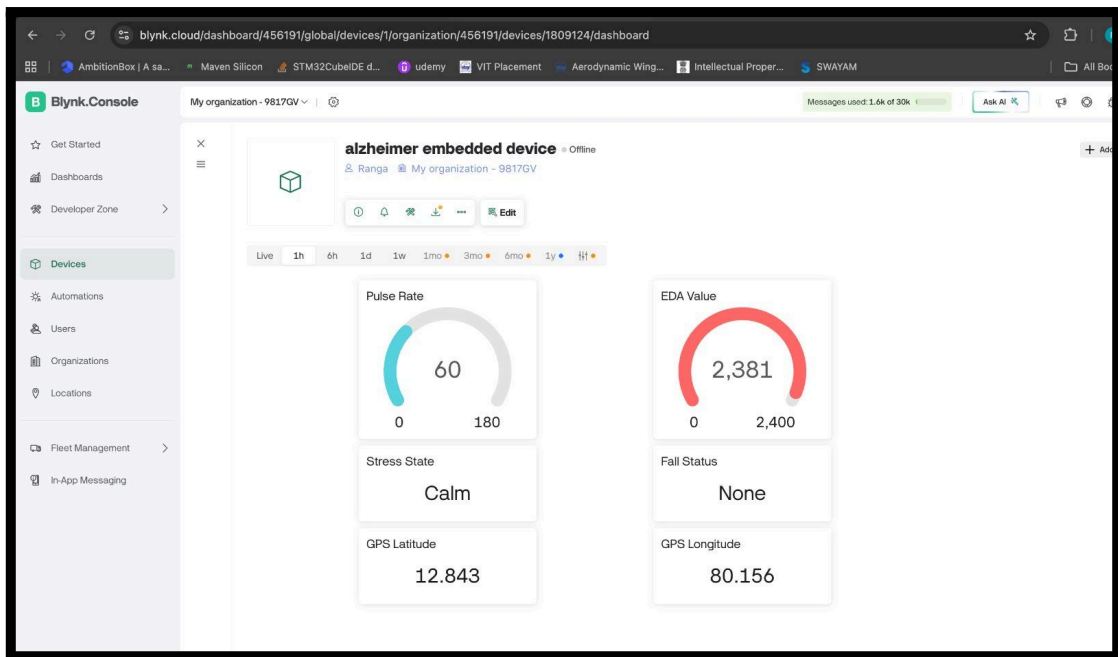


Figure 3: Reading of real time data Blynk website

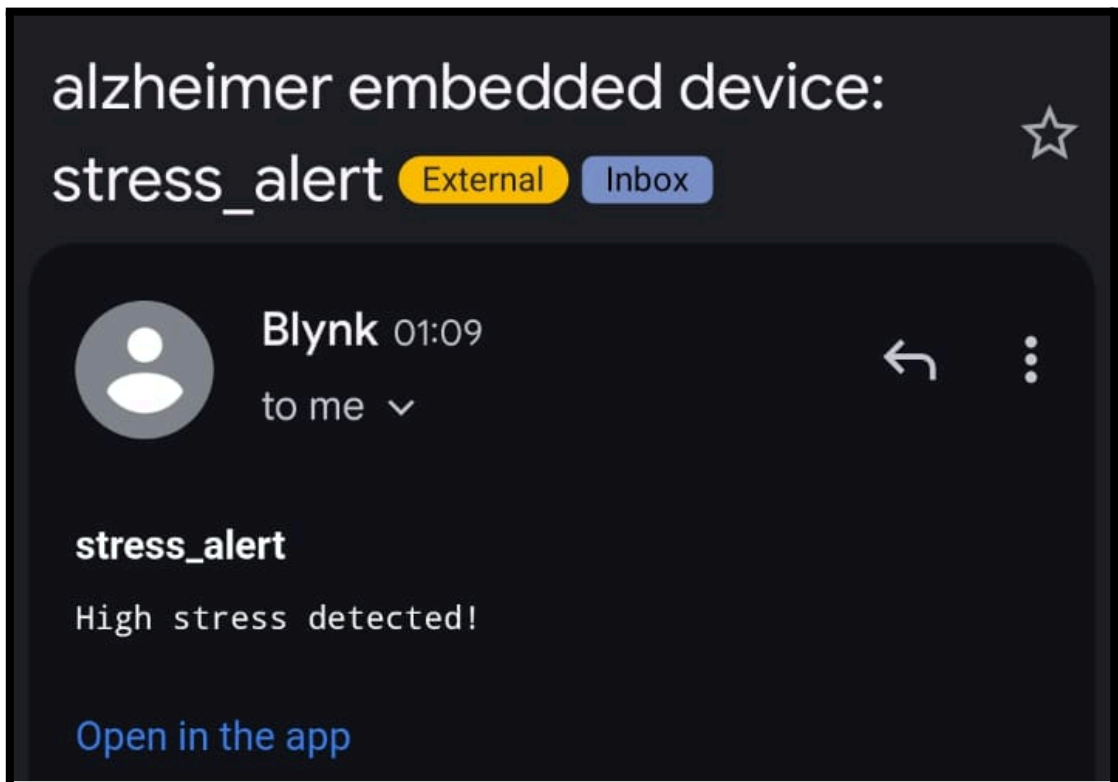


Figure 4: Real time data alert Blynk website for mail

CHAPTER V

5. CONCLUSION

5.1 COST ANALYSIS

The proposed Embedded Device for Alzheimer's Patients was developed as a low-cost and efficient prototype using readily available components. The total cost of implementation remains affordable, making it practical for home-based healthcare assistance. The main expenses involve the sensors, ESP32 microcontroller, and supporting modules.

Component	Quantity	Approx. Cost (INR)
ESP32 Microcontroller	1	₹450
NEO-6M GPS Module	1	₹600
Pulse Sensor	1	₹250
MPU-6050 Sensor	1	₹800
GSR Sensor	1	₹300
DFPlayer Mini + Speaker	1	₹350
Power Supply / Battery	1	₹300
Connecting Wires, PCB, Misc. Components	-	₹250
Total Estimated Cost		₹3,300 – ₹3,500

Table 4: Cost analysis and approximation

This cost analysis shows that the system is economically feasible and suitable for large-scale production with minor modifications.

5.2 SCOPE OF WORK

The scope of this project is to design, develop, and test an embedded assistive device capable of improving safety and quality of life for Alzheimer's patients.

Current Features Implemented:

- Real-time location tracking using GPS (NEO-6M).
- Continuous heart rate monitoring using Pulse Sensor.
- Fall detection using MPU-6050 accelerometer.
- Stress monitoring with GSR Sensor.
- Voice-based task reminders through DFPlayer Mini + Speaker.
- ESP32 microcontroller integration for processing and control.

Future Scope:

- Integration with mobile applications for caregiver alerts.
- Cloud storage for long-term health data tracking.
- Use of AI algorithms for predicting falls or health risks.
- IoT-based notifications for remote monitoring via the internet.
- Compact wearable version for continuous patient use.

5.3 SUMMARY

The project successfully demonstrates the design and implementation of an embedded device for Alzheimer's patients that combines multiple sensors for health monitoring and safety management. Each module:- GPS, Pulse Sensor, MPU-6050, and GSR Sensor, works in coordination with the ESP32 microcontroller to collect, process, and analyze data in real time.

The audio reminder system ensures that patients follow their medication and daily routines, while caregivers receive timely updates on patient status. The project meets

its objectives by providing a cost-effective, user-friendly, and reliable solution to assist Alzheimer's patients in maintaining independence and safety.

Overall, this system represents a step toward smart healthcare monitoring, bridging the gap between human care and technology. With further enhancements such as mobile integration and AI-based analytics, the device has strong potential for real-world medical and home-care applications.

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